

The empirical recurrence for OEIS A299331, recovered from its tail

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Abstract

OEIS A299331 records “Empirical recurrence of order 76” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 119$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 80$. Its last coefficient is $c_{76} = -107200 \neq 0$, so no further tail satisfies anything shorter; any order-76 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299331 is “Number of $n \times 5$ 0..1 arrays with every element equal to 2, 3, 4, 5, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 17, 173, 2537, 46286, 816886, 14783424, 267652693, ...

The entry states:

Empirical recurrence of order 76 (see link above)

As of the “Last modified” line on the live entry (Feb 07 2018 04:49:18, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 119$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 119$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 76: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 76.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 238$ exact terms from $n = 4$ on, it returns order 76 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{76} c_i a(n-i),$$

c_1	13	c_{20}	−34827786166	c_{39}	−81625708886375	c_{58}	−10551195937577
c_2	163	c_{21}	−54951688190	c_{40}	−2302008359892	c_{59}	−2818811546955
c_3	−669	c_{22}	57418385291	c_{41}	112050201159377	c_{60}	4934587675485
c_4	−10725	c_{23}	139677933712	c_{42}	−132321358943824	c_{61}	−148605850059
c_5	−11854	c_{24}	228758485867	c_{43}	43948965650275	c_{62}	118277915250
c_6	208247	c_{25}	760192641290	c_{44}	20709204403527	c_{63}	−337648588409
c_7	758867	c_{26}	−631730359446	c_{45}	−56912139500731	c_{64}	−376879087943
c_8	123202	c_{27}	794027025790	c_{46}	41272656076890	c_{65}	163956681377
c_9	−5858538	c_{28}	−1058582943142	c_{47}	−7271589977228	c_{66}	9435075191
c_{10}	−26021532	c_{29}	−4291807741500	c_{48}	27537712461529	c_{67}	−32429640303
c_{11}	−42080490	c_{30}	5402711712340	c_{49}	−4869300071107	c_{68}	13269258493
c_{12}	−5834236	c_{31}	−9506536098416	c_{50}	−34437025886727	c_{69}	6258456317
c_{13}	304867080	c_{32}	6965260646496	c_{51}	27791378990691	c_{70}	−1111596678
c_{14}	1178053573	c_{33}	14272258958391	c_{52}	16873175961664	c_{71}	−576427164
c_{15}	1837546243	c_{34}	−23948038193988	c_{53}	−34382069598162	c_{72}	−14856856
c_{16}	1304555358	c_{35}	37649279920731	c_{54}	24012739299810	c_{73}	14027130
c_{17}	−2181459106	c_{36}	−13583065144355	c_{55}	−19576136129451	c_{74}	6202232
c_{18}	−13823401253	c_{37}	−39717378466274	c_{56}	−1494525207345	c_{75}	498952
c_{19}	−32674347969	c_{38}	78970933738082	c_{57}	17469732554604	c_{76}	−107200

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 80$, and it is the recurrence OEIS A299331 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{76} - c_1 x^{75} - \dots - c_{76}$. Its constant term is $-c_{76} = 107200$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 76 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 76, so $\deg q = 76$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 76, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 76, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -107200 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-76 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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